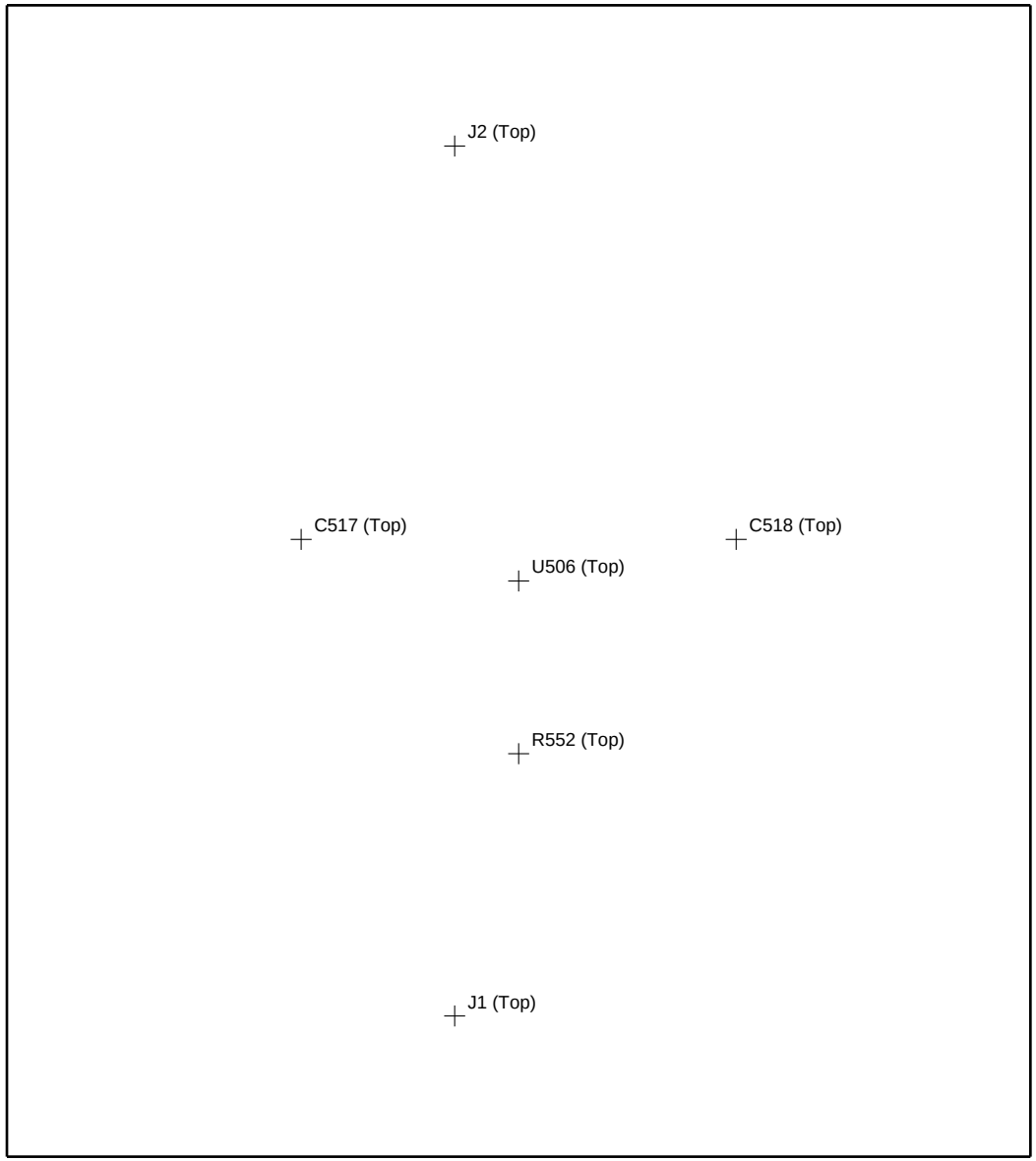

Board extent 16.000 mm (X) x 18.000 mm (Y)

Top view, Edge.Cuts and holes of HDMI_INSERT.kicad_pcb; crosses = CPL positions (KiCad footprint origins). Not to a fixed scale.
NOT RELEASED: model/CI only; bench gates OPEN

HDMI_INSERT fab notes

NOT RELEASED: model/CI only; bench gates OPEN.
Board status: SB2_ROUTED_MECH_FIT_OPEN_NOT_FOR_FABRICATION. Every value below is a CI/model output of pcb/fab_export.py and pcb/board_gen.py; no hardware or bench gate is closed by it.

Board

Outline (Edge.Cuts of HDMI_INSERT.kicad_pcb): 16.000 x 18.000 mm, area 288.0 mm2; lower-left corner at X 100.000, Y -118.000 mm in the Gerber/CPL frame (KiCad absolute origin, Y up).
Holes: no mounting holes.

Fabrication

Copper layers: 2; finished thickness 1.6 mm (JLC tolerance +/-10 % for boards >= 1.0 mm).
Stack-up: JLC standard 2-layer FR-4, outer copper 1 oz (https://jlcpcb.com/capabilities/pcb-capabilities, read 2026-11-07).
Impedance: no RF50 net on this board; order without impedance control.
Surface finish, solder mask and legend colours: order options, not fixed by this fab set.
KiCad 10.0.7 DRC on the committed board: 0 errors, 0 unconnected, 4 reviewed warnings, ignored checks [] (report.json; CI/model result).

Files (fab/)

gerber/: Gerber X2 with netlist attributes, Protel extensions, one file per layer, Gerber job file (.gbrjob). drill/: Excellon PTH and NPTH files (mm, decimal, absolute origin, alternate oval mode) and Gerber X2 drill maps. HDMI_INSERT.ipc356: IPC-D-356 netlist of every net with a pad. assembly/: JLC BOM and CPL, the raw KiCad position file, the rotation offset table and the instance map.

Assembly

Boards to build: 1 = instances (HDMIA 1) + spares 0; every board is the same design and is assembled the same.
Sides with parts: Top (SMD 4, THT 2).
CPL: Mid X / Mid Y are the KiCad footprint origins (JLC defines them as the component centroid) and Rotation the raw KiCad rotation (degrees, counter-clockwise as the JLC CPL column). JLC rotation and centroid offsets per footprint are TBD (assembly/HDMI_INSERT-cpl-rotation-offsets.csv): check every part in the JLC viewer at order time; nothing here verifies them.
BOM TBD: J1 (FPC socket A placeholder (pin 18 in, GND), nothing fitted): TBD: pilot_placeholder, no graph ref and no MPN yet; replaced by FPC socket A for flex 1 (MPN TBD; W2 GR-4 OPEN): its pin-18 and GND contacts
BOM TBD: J2 (FPC socket B placeholder (pin 18 out, GND), nothing fitted): TBD: pilot_placeholder, no graph ref and no MPN yet; replaced by FPC socket B for flex 2 (MPN TBD; W2 GR-4 OPEN): its pin-18 and GND contacts

Not proved here (OPEN)

- that the schematic is right
- current capacity or temperature rise beyond the chosen class widths
- impedance, RF loss or isolation, EMC or signal integrity
- creepage beyond the encoded clearances
- that the enclosure fits (only outline and hole checks; heights are not checked here)
- that the fab's DFM accepts the board, or that CPL rotations are right
- that the land patterns match the purchased MPNs
- any function: hardware and bench gates stay OPEN

Sources (read 2026-11-07): https://jlcpcb.com/help/article/bill-of-materials-for-pcb-assembly; https://jlcpcb.com/help/article/pick-place-file-for-pcb-assembly; https://jlcpcb.com/help/article/how-to-generate-the-bom-and-centroid-file-from-kicad; https://jlcpcb.com/help/article/how-to-generate-gerber-and-drill-files-in-kicad-8; https://jlcpcb.com/capabilities/pcb-capabilities